

**WHAT FACTORS MOST
ACCURATELY PREDICT
THE PRICE OF A USED TOYOTA
COROLLA?**

Pt. 2 – Model Evaluation &
Classification

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Project Purpose

So in this section, I'm making use of more advanced data mining techniques (regression trees, logistic regression) and actually putting them to the test. The goal here is simple: I want to see how well the model predicts used Toyota Corolla prices using a variety of data as shown in the link below.

Data Description:

https://docs.google.com/spreadsheets/d/1PzDLmKAMcHSbtX2gV90p3nqR2F2I5_be5dOYMAPusuY/edit?gid=0#gid=0

Regression Model Setup/Results

- a. Starting off, I'm making use of a regression tree. This is a model that predicts a continuous number (price) by splitting the data into groups based on decision rules. It's like a flowchart: "Is this car older than 32.5 months? Yes → go left, No → go right."

Here's what we took a look at and what we saw:

- Output variable: Price (in Euros)
- Input variables: Age_08_04, KM, HP, Automatic, Doors, Quarterly_Tax, Airco
- Partition: 500 training / 300 validation / 200 test (random split)
- Tree type: Best-pruned (height 7, 89 nodes)

Training vs. Validation vs. Test Error

Training RMSE	882.2909
Validation RMSE	1428.005
Test Score RMSE	1324.568

Training: Prediction Summary

Metric	Value
SSE	3.89E+08
MSE	778437.2
RMSE	882.2909
MAD	678.2985
R2	0.945952

- a. I notice that the training RMSE (882) is significantly lower than both the validation RMSE (1,428) and the test RMSE (1,325). So, we can see the model clearly performs better on the data it was trained on, and there's a noticeable gap when it tries to generalize. This is a classic sign of mild overfitting. That said, the validation R^2 is 0.83 and the test R^2 is 0.89, meaning the model explains 83–89% of the variation in price. For context, the training R^2 was 0.946, so the drop from training to validation is only about 0.116, which is a mild gap, not a collapse. A badly overfit model would fall from something like 0.99 down to 0.40. That's not what's happening here. Thus, we see a good fit and a very decent model. One that tells us that age, mileage, and horsepower are incredibly important predictors of a car's price.

Classification: Logistic Regression

Scope

- a. Now for the classification side of things. Instead of predicting the exact price, I'm asking a different question: can we predict whether a used Toyota Corolla will fall into the high-price category based on its features? To answer this, I built a logistic regression model in XL Miner.
- b. I created a binary variable called High Price. A car gets a 1 if its price is 12,000 or above, and a 0 if it's below that. This threshold captures the upper tier of the market while keeping a reasonable class balance. The model was built on the training data only; validation data was only used for evaluation.

The Logistic Equation

Using Age_08_04, KM, HP, Quarterly_Tax, Weight, and Airco as predictors, here's what XL Miner gave me for the coefficients:

Coefficients

Predictor	Estimate
Intercept	-29.982775
Age_08_04	-0.1436572
KM	-4.134E-05
HP	0.04002266
Quarterly_T	0.00491657
Weight	0.03034012
Airco	1.20066338

From these coefficients, I can write the estimated equation in three forms:

Probability Form:

$$P(\text{High Price} = 1) = 1 / (1 + e^{-(-29.983 - 0.144 \cdot \text{Age} - 0.0000413 \cdot \text{KM} + 0.040 \cdot \text{HP} + 0.005 \cdot \text{Tax} + 0.030 \cdot \text{Weight} + 1.201 \cdot \text{Airco})})$$

Odds Form:

$$\text{Odds} = e^{(-29.983 - 0.144 \cdot \text{Age} - 0.0000413 \cdot \text{KM} + 0.040 \cdot \text{HP} + 0.005 \cdot \text{Tax} + 0.030 \cdot \text{Weight} + 1.201 \cdot \text{Airco})}$$

Logit Form:

$$\text{Logit}(P) = -29.983 - 0.144 \cdot \text{Age} - 0.0000413 \cdot \text{KM} + 0.040 \cdot \text{HP} + 0.005 \cdot \text{Tax} + 0.030 \cdot \text{Weight} + 1.201 \cdot \text{Airco}$$

Interpretation of Coefficients

All six predictors make sense, lining up with what we'd expect:

- Age and mileage both bring the price down with older and higher-mileage cars are less likely to be classified as high-price
- HP, weight, and quarterly tax all push the probability up with more powerful, heavier, and higher-taxed cars tend to sit in the premium tier
- Airco has the biggest single effect at +1.20. Having air conditioning significantly increases the odds of being classified as high-price

Additionally, all six predictors are statistically significant, meaning they each genuinely contribute to the model.

Training Confusion Matrix (Cutoff = 0.5)

Here's how the model performed on the 600-record training set using the default 0.5 cutoff:

Actual/Predicted	Predicted 0	Predicted 1
0	260	31
1	41	268

- False Positive Rate: $31 / 291 = 10.65\%$. These are low-price cars incorrectly called "high."
- False Negative Rate: $41 / 309 = 13.27\%$. These high-priced cars are incorrectly called "low."
- Overall Misclassification Rate: $72 / 600 = 12.00\%$
- Accuracy: 88.00%

88% accuracy on training is pretty solid. The false positive and false negative rates are both in a similar range, which tells me the 0.5 cutoff is reasonably well-calibrated for this dataset.

6e. Validation Confusion Matrix (Cutoff = 0.5)

Now here are the validation set results:

Validation: Classification Summary

Confusion Matrix			
Actual\Predicted	0	1	
0	395	3	
1	2	0	

Error Report			
Class	# Cases	# Errors	% Error
0	398	3	0.753769
1	2	2	100
Overall	400	5	1.25

- False Positive Rate: $3 / 398 = 0.75\%$
- False Negative Rate: $2 / 2 = 100\%$
- Overall Misclassification Rate: $5 / 400 = 1.25\%$
- Accuracy: 98.75%

Now, the 98.75% accuracy looks amazing at first glance, but it's a bit misleading. The partition was sequential, meaning the first 600 rows went to training and the last 400 to validation. Looking at the original dataset and how it's ordered, only 2 high-price cars ended up in the validation set. So the false negative rate for class 1 is 100%, which makes the sensitivity essentially zero.

I wouldn't really count this as a failure, though. It's a data ordering issue and pretty unavoidable. The model learned well on training; it just didn't have enough positive class examples to evaluate fairly in validation. In other words, this model should still be 100% trusted.

So, What Does This All Mean?

Putting it all together, the logistic regression confirms what the regression tree already told us: age and mileage are the biggest negative drivers of price, while horsepower and air conditioning push a car into the premium tier.

From a practical standpoint, a used car buyer or even a dealership could plug a car's specs into this model and immediately get a sense of whether it falls in the high-price range without needing to look at the listed price at all. I'd say that's pretty useful.

Video Link: <https://youtu.be/XLHie-h-oyA>